

Physics Aware Multi-Task Learning for Forest and Biomass Estimation

Biomass and forest height are non-linearly related to each other through an allometric equation, which describes the relationship between size of tree species to various other parts of the tree often in the form of a mathematical equation or a model. It is expressed as a power law where one biological variable (Y) scales another variable (X) according to the equation $Y = aX^b$ where a and b are scaling coefficients.

Above Ground Biomass (AGB) of a single tree can be expressed as a product of tree volume (tree species specific) and wood density. The tree volume is in turn estimated using the stem diameter at breast level (DBH), tree height, and tree shape factor. With the spatial resolution available with satellite imagery, single tree biomass estimation is not possible. In such cases the single tree estimations are replaced with mean stand biomass estimation.

Objective: To establish a deep learning model using the multitask learning approach to jointly estimate forest height and biomass while learning shared representations that help each other in increasing prediction accuracy while reducing the need for field measurements to estimate biomass and increase reliability on model predictions.

Methodology

1. Dataset Availability

Utilize existing datasets like LiDAR, Synthetic Aperture Radar such as Tandem-X satellite images, and field measurements from forest inventories. For reference target, the AfriSAR dataset offers grided forest characterization products such as canopy height and above ground biomass (AGB) derived from full waveform LiDAR data over five forested sites such as Lope, Mondah, Pongara, Rabi, and Mabouni in Gabon, Africa. Field measurements that include the tree species are available in the “AfriSAR: Mondah Forest Tree Species, Biophysical, and Biomass Data, Gabon, 2016” dataset. These datasets provide forest height, AGB, and tree species for a deep learning based estimation.

2. Model Development

Multitask Learning Framework: Develop a deep learning model using a multitask learning approach where the model will have i) two outputs: forest height and tree species as depth estimation head and segmentation head respectively which is in turn used for AGB estimation or ii) three outputs: forest height, tree species, and AGB estimation. The shared layers of the model will learn common representations from the data while task specific layers learn to optimize for forest height, tree species, or biomass.

Integration of Allometry: Introduce the Forest-Biomass allometric equation to the model as a physics constraint or a regularizing term in the form of a loss function. This ensures that the predictions adhere to biologically plausible relationships between height and biomass. Loss function here is a combined loss function that includes a task specific loss such as cross-entropy for segmentation head and mean squared error for regression task. A weighted sum of the two task specific losses is used to obtain the multitask loss with the options of constant weight factors or dynamic weight factors calculated via Uncertainty-Based Weighting (assigns higher weights to tasks that are more uncertain or have higher losses) or GradNorm approach (dynamically adjust the weights based on the gradients of each task's loss, ensuring that all tasks contribute equally to the learning process). Normalization of losses is required if the losses are in different magnitude scales.

Training and Validation: Train and validate the model using dataset splits and evaluate using performance metrics such as RMSE for forest height and biomass estimation and Intersection over Union (IoU) and the Dice Coefficient for tree species segmentation.

3. Model Evaluation

Compare multitasking learning performance with a single task model that estimates forest height and biomass separately. Check for generalization capability in different regions with similar forest characteristics.

Outcome & Significance

- The proposed model is expected to deliver more accurate and efficient estimates of forest height and biomass compared to traditional methods and single-task models.
- The integration of allometric principles into deep learning models represents a novel approach that bridges traditional ecological knowledge with modern computational techniques.
- Reliability on the estimates of forest height and biomass is increased with the inclusion of an allometric equation which improves confidence for downstream usage in decision making, monitoring, and conservation efforts.